

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2011

Dewey 138 kV
January 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2011. The Interconnection Customer proposes a 230.2 MW natural gas combined cycle facility interconnecting at the Dewey 138 kV substation in the Kentucky area. The request was received on February 19, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 710 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$210,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Carlton Hills Substation – Orovada 138kV Line is overloaded to 102.5% of its 463 MVA emergency rating for the loss of the Carlton Hills – Atoka 345kV. The project contributes 5.4% DFAX (12.5 MW) to this overload.

Carlton Hills 138/1kV XFMR: 100.6% loading (481 MVA) following the outage of the Atoka – Carlton Hills Substation 345kV; DFAX = 5.4%.

The Ragland – Butler Ridge 138kV line is overloaded to 100.5% of its 296 MVA emergency rating for the loss of the Neptune – Ragland 345kV L. The project contributes 6.8% DFAX (15.7 MW) to this overload.

The Cinder – Butler Ridge 138kV is overloaded to 100.3% of its 304 MVA emergency rating for the loss of the Ragland – Neptune 345kV. The project contributes 6.8% DFAX (15.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carlton Hills Substation – Orovada 138kV Line	Reconductor line	\$195,000	3% (\$5,000)
Carlton Hills 138-1 kV Auto	Replace transformer with higher-rated unit	\$4,951,000	3% (\$127,000)
Ragland Substation – Butler Ridge 138kV	Reconductor line	\$748,000	3% (\$25,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cinder – Butler Ridge 138kV line	Reconductor line	\$1,550,000	3% (\$53,000)
	Total allocated to project		\$210,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.